

Three-phase Controlled Rectifiers

Need?

Comparison with single phase!

Features of three-phase controlled rectifiers

- Operate from three-phase ac supply voltage.
- They provide higher dc output voltage.
- Higher dc output power.
- Higher output voltage ripple frequency.
- Filtering requirements are simplified for smoothing out load voltage and load current.

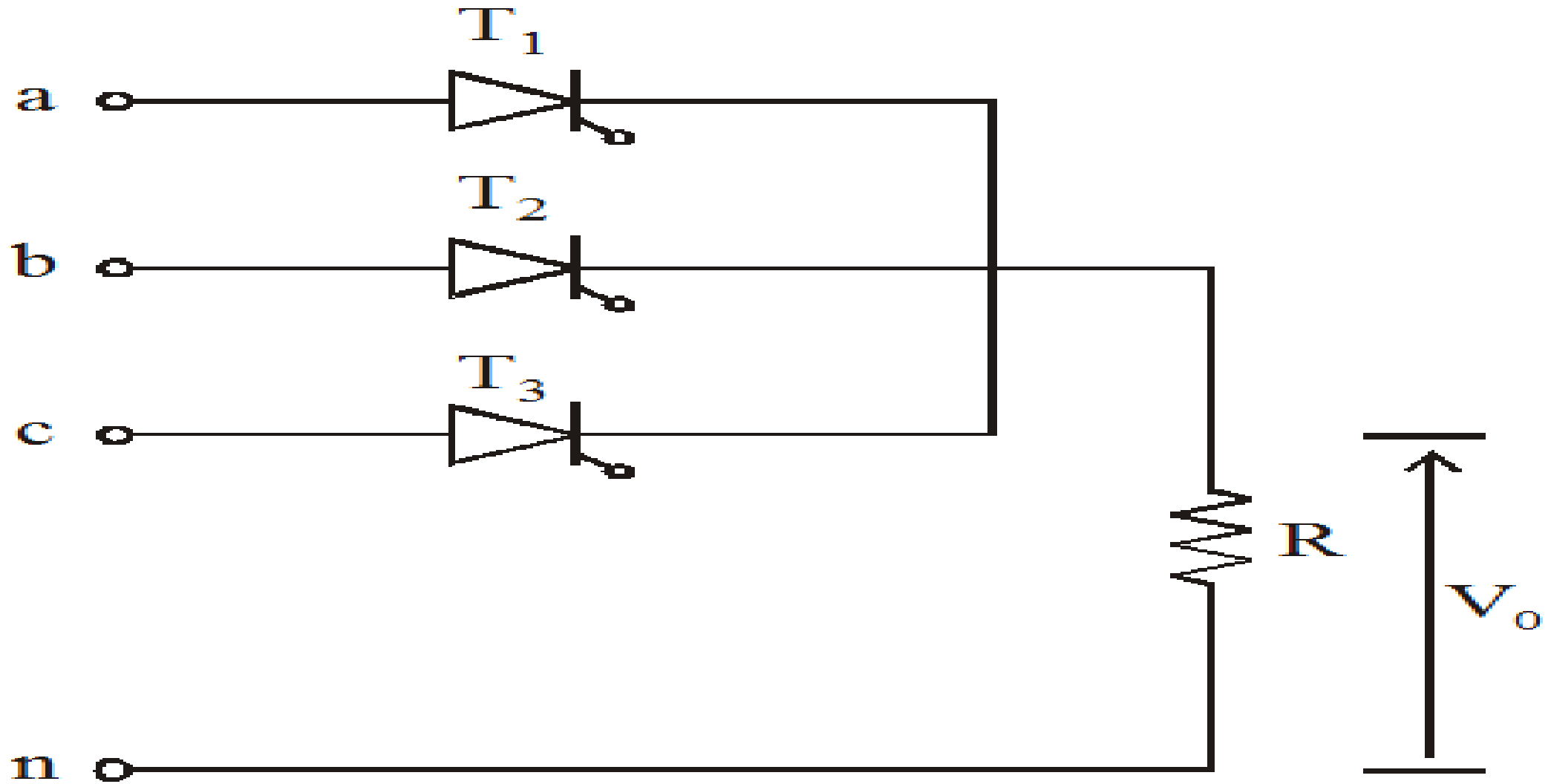
Classification of 3-phase converters

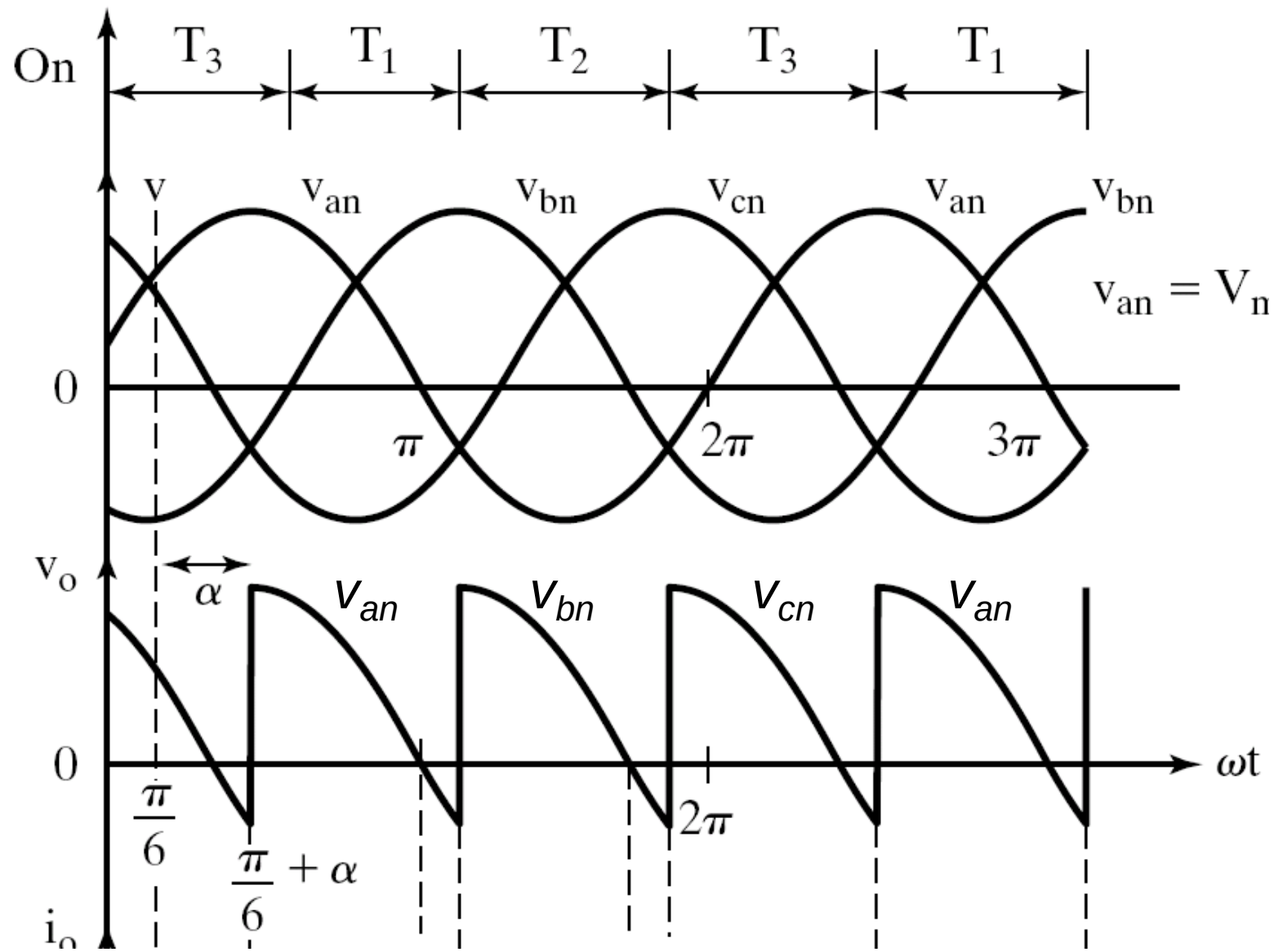
- 3-phase half wave converter
- 3-phase semi converter
- 3-phase full converter
- 3- phase dual converter

3-Phase Half Wave Converter (3-Pulse Converter)

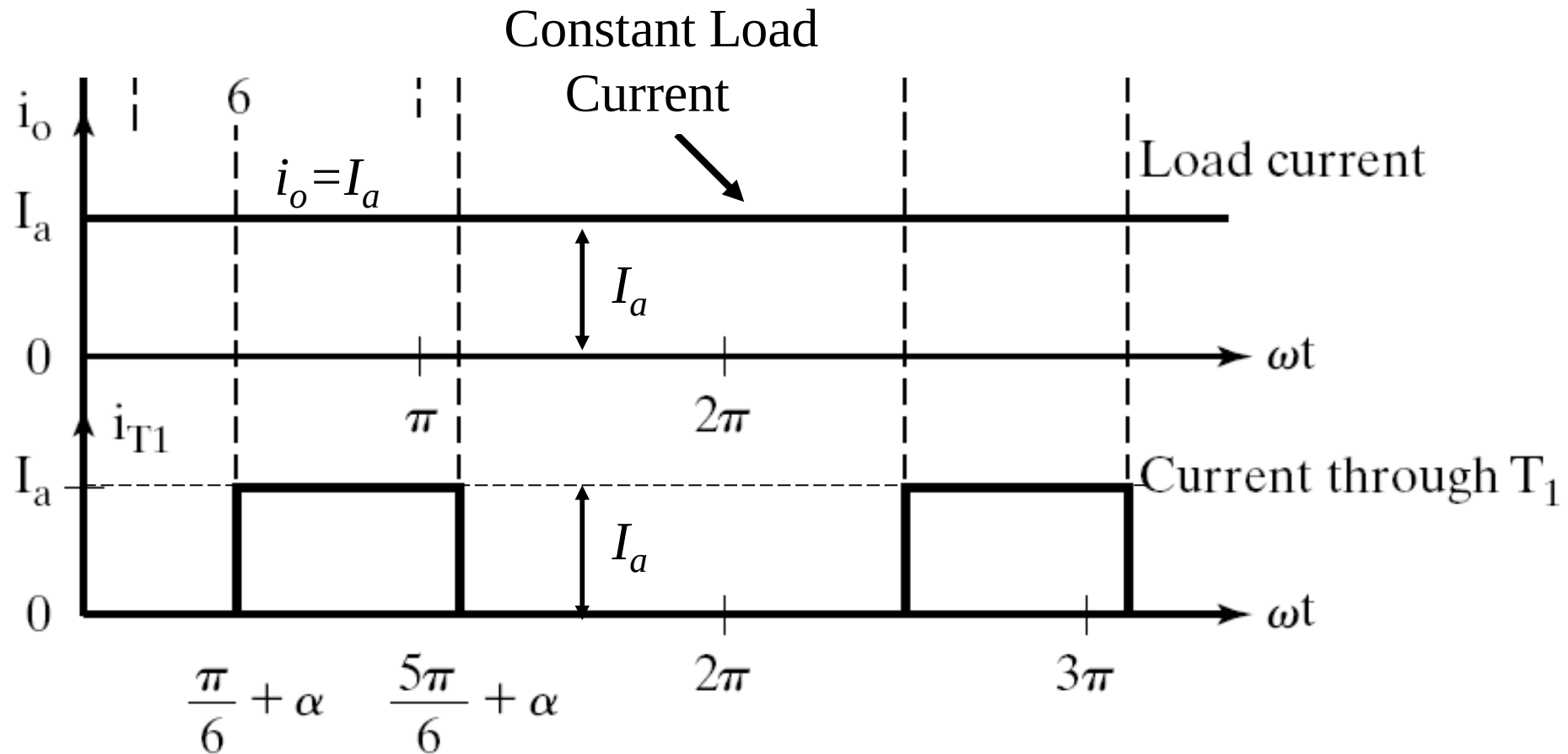
Points of natural commutation and α

Circuit Diagram





Each thyristor conducts for $2\pi/3$ (120°)



T_1 is triggered at $\omega t = \left(\frac{\pi}{6} + \alpha \right) = (30^\circ + \alpha)$

T_2 is triggered at $\omega t = \left(\frac{5\pi}{6} + \alpha \right) = (150^\circ + \alpha)$

T_3 is triggered at $\omega t = \left(\frac{9\pi}{6} + \alpha \right) = (270^\circ + \alpha)$

Each thyristor conducts for 120° or $\frac{2\pi}{3}$ radians

If the reference phase voltage is

$v_{RN} = v_{an} = V_m \sin \omega t$, the average or dc output voltage for continuous load current is calculated using the equation

$$V_{dc} = \frac{3}{2\pi} \left[\int_{\frac{\pi}{6} + \alpha}^{\frac{5\pi}{6} + \alpha} V_m \sin \omega t \cdot d(\omega t) \right]$$

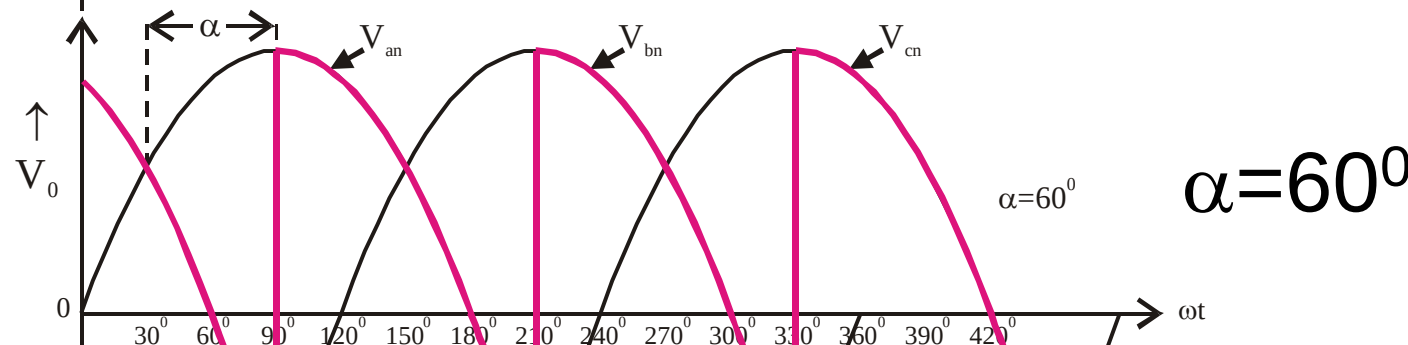
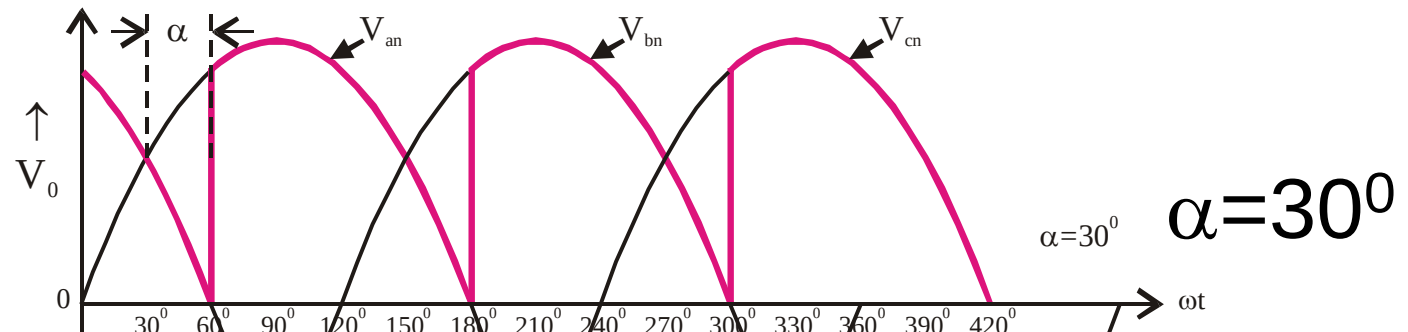
$$V_{dc} = \frac{3\sqrt{3}V_m}{2\pi} \cos(\alpha)$$

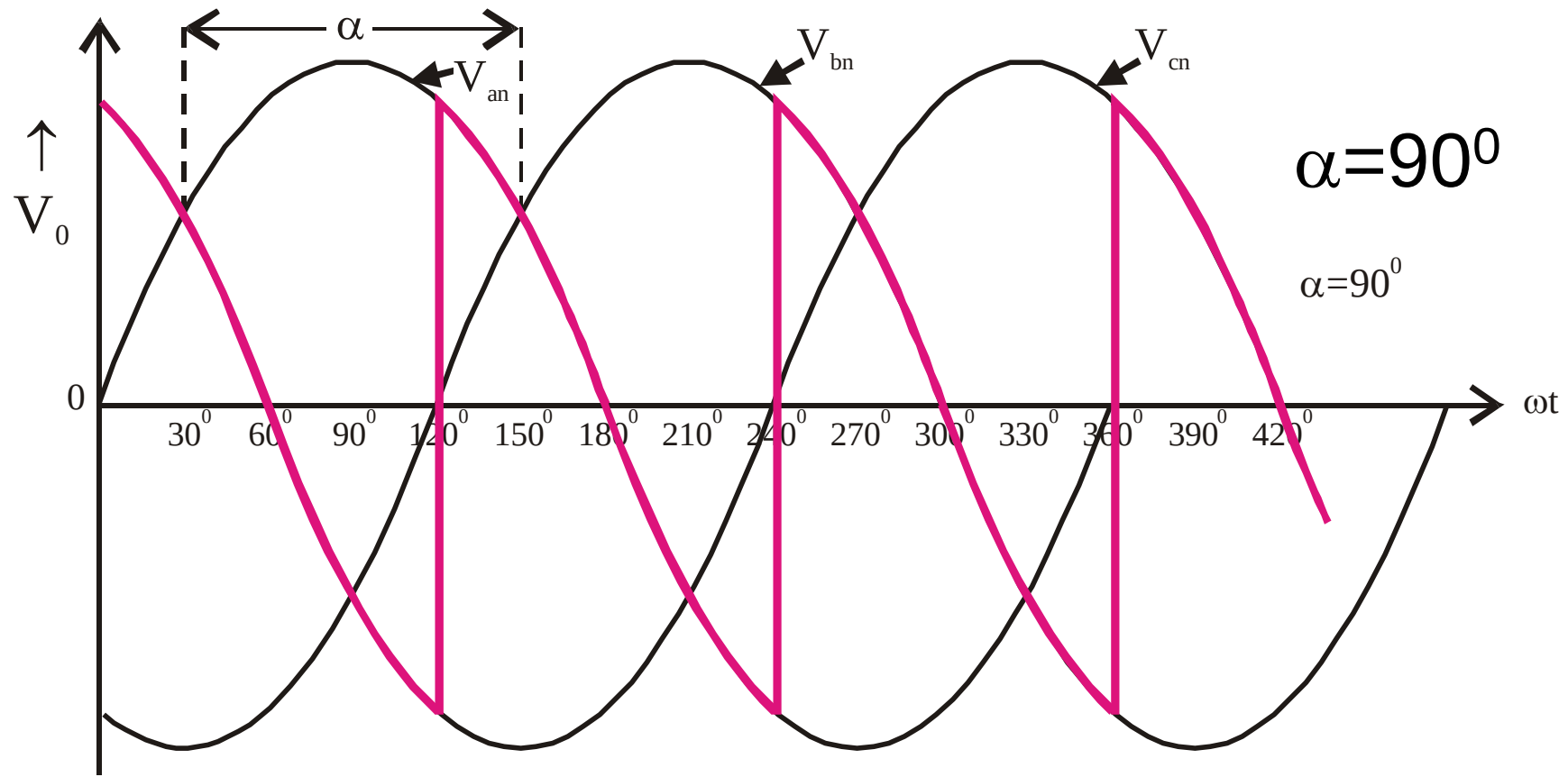
OR

$$V_{dc} = \frac{3V_{lm}}{2\pi} \cos(\alpha)$$

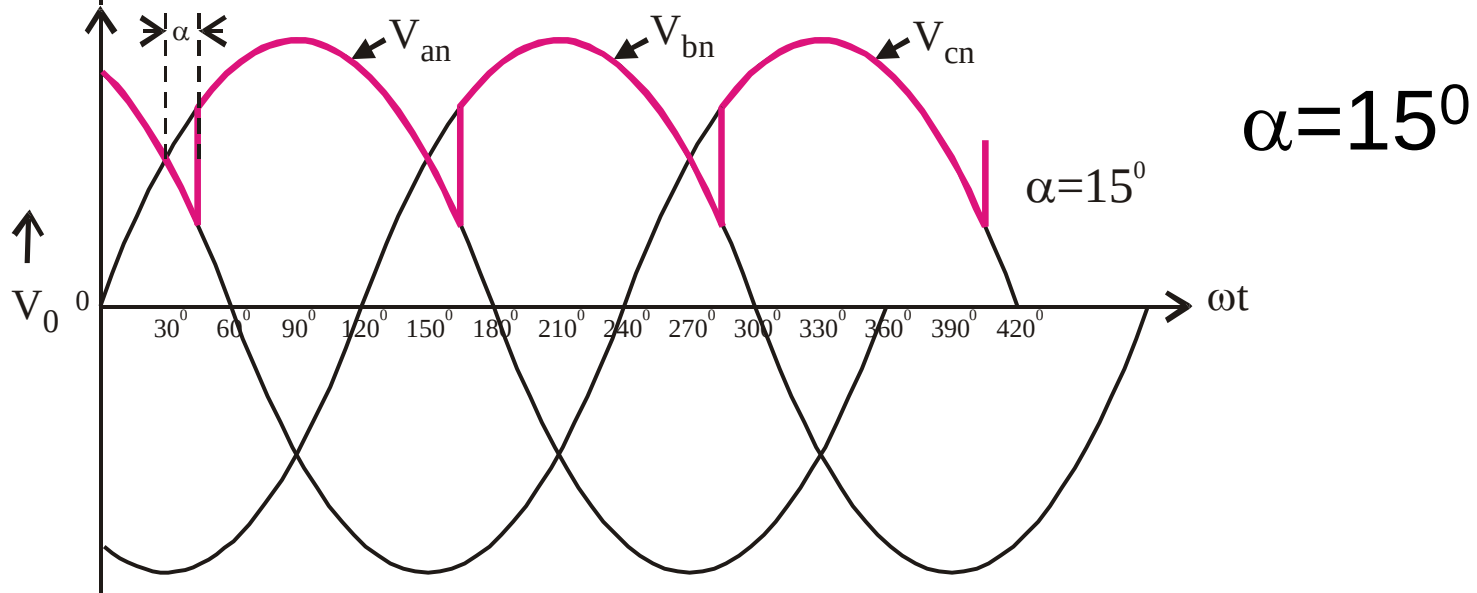
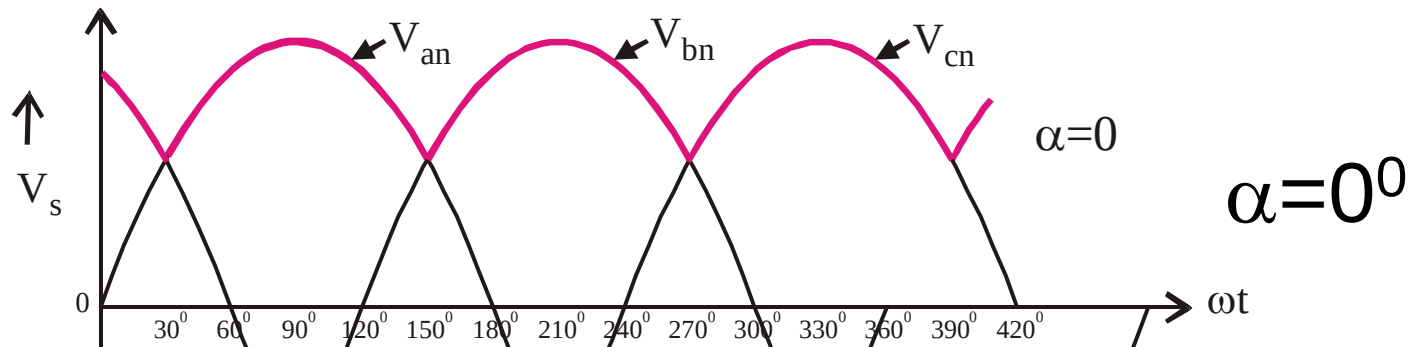
Where $V_{lm} = \sqrt{3}V_m =$ Max. line to line supply voltage

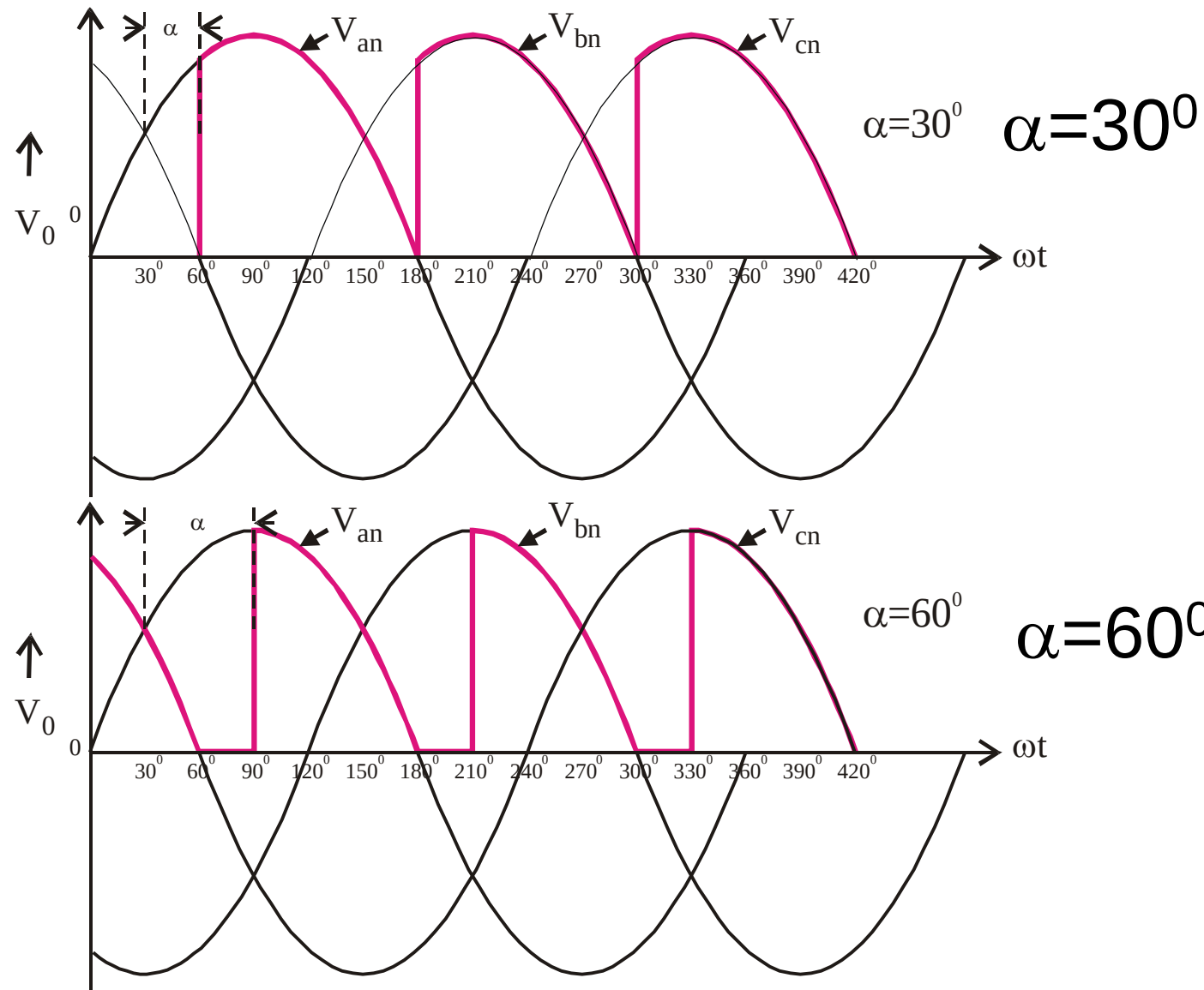
Three Phase Half-Wave Controlled Rectifier Output Voltage Waveforms For RL Load at Different Trigger Angles





Three-Phase Half Wave Controlled Rectifier With R Load and RL Load with FWD



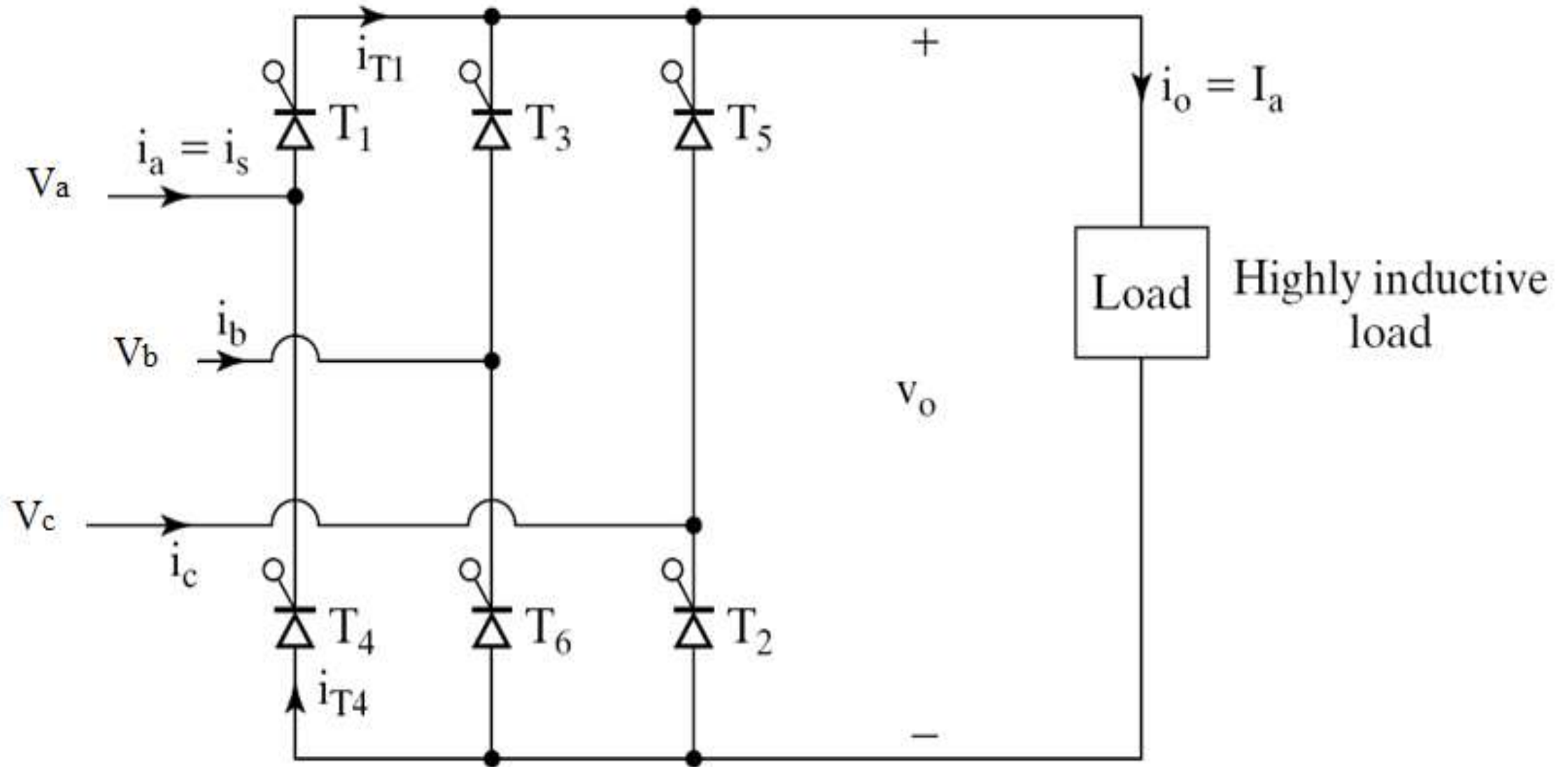


Three Phase Full Converter

Three Phase Full Converter

- 3 Phase Fully Controlled Full Wave Bridge Converter.
- Known as a 6-pulse converter.
- Used in industrial applications up to 120 kW output power.
- Two quadrant operation is possible.

Circuit Diagram



Waveforms for CCM and $\alpha = \pi/3$

